

Problem - Solution Fit

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Problem - Solution Fit Canvas:

The **Human Development Index (HDI)** is a composite measure of a country's development based on **life expectancy, education, and Gross National Income (GNI) per capita**. Analyzing these indicators manually can be time-consuming and may not provide quick insights into a country's development status. This project develops a **Machine Learning-based Human Development Index (HDI) Predictor** that analyzes key socio-economic indicators to predict the HDI score and classify countries into **Very High, High, Medium, or Low** human development categories. The trained model is integrated into a **Flask web application**, enabling users to enter development indicators and receive instant, accurate HDI predictions. The system supports policymakers, researchers, governments, and development organizations in evaluating human development and making informed policy decisions through a scalable web-based solution.

Purpose:

- Develop an intelligent **Machine Learning-based HDI Prediction System** using life expectancy, education, and GNI per capita indicators.
- Perform **data preprocessing, exploratory data analysis (EDA), visualization, and feature engineering** to improve prediction accuracy.
- Train and evaluate a **Linear Regression** model to accurately predict the Human Development Index (HDI) score.
- Build a **user-friendly Flask web application** that allows users to input country development indicators and receive instant HDI predictions.
- Support **governments, policymakers, researchers, and international organizations** in assessing development levels and identifying areas requiring improvement.
- Save the trained model using **Pickle serialization** and integrate it into the web application for real-time predictions without retraining.
- Deploy the solution as a scalable web application to provide **accessible, reliable, and real-time Human Development Index predictions** for development planning and decision-making.

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? • Governments & Policymakers • Researchers & Academicians • International Organizations (UNDP, World Bank, WHO) • Development Agencies & NGOs • Students & Data Science Learners • General Public & Data Enthusiasts	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? (i.e. limited data availability, complex indicators, lack of technical expertise, time constraints, cost, limited tools, etc.) • Limited access to updated and reliable HDI data • Lack of expertise in data analysis and ML models • Complex calculation of HDI indicators manually • Time-consuming data processing and analysis • Cost of advanced analytics tools • Need for an easy-to-use and accessible solution	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they used in the past? What pros & cons do these solutions have? (i.e. traditional reports, manual calculations, basic dashboards, existing tools, etc.) • UNDP HDI reports and databases • Excel-based manual calculations • Basic dashboards & static reports • Online HDI calculators • Data portals like World Bank Open Data • Pros: Reliable but not real-time, Cons: Not interactive or predictive	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which job-to-be-done (or problems) do your customers try to solve? • Predict the Human Development Index (HDI) of countries • Analyze the impact of life expectancy, education, and income on development • Identify development gaps in countries • Compare human development across countries • Support data-driven policy and decision making • Monitor progress toward development goals	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? (i.e. root causes that trigger the problem) • Lack of integrated analysis of key development indicators • Limited understanding of how indicators affect HDI • Time-consuming manual calculations and comparisons • Insufficient visualization and insights • Delayed decisions due to lack of real-time predictions • Limited access to user-friendly predictive tools	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? (i.e. how do they currently try to solve it? Their habits, workarounds, existing efforts) • Use published reports and static data sources • Perform manual calculations in spreadsheets • Rely on academic research and studies • Use basic data visualization tools • Seek expert help for analysis • Make policy decisions with limited insights	Focus on J&P, tap into BE, understand RC
Identify TR & EM	3. TRIGGERS TR What triggers customers to act? (i.e. noticing development gaps, new data release, policy planning, academic research, report deadlines, global rankings, etc.) • Release of new HDI data or reports • Need to evaluate country development status • Policy planning and decision-making • Academic research and studies • Global rankings and comparisons • Development goals and target tracking	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first. If in the new venture, define how your HDI Predictor is a solution. (i.e. how our ML-based system analyzes indicators, predicts HDI, provides insights, and supports decision making.) • Machine Learning model predicts HDI score using life expectancy, education, and GNI per capita. • Linear Regression model ensures accurate and reliable predictions. • Data preprocessing and feature engineering improve accuracy. • Interactive Flask web application provides instant predictions. • Visualizations and insights help identify development gaps. • Deployed web app offers real-time, accessible HDI predictions for all users.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 • Access web application • Enter indicators and get predictions • View visualizations and reports • Download or share insights • Access datasets and documentation 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. • Attend workshops and training • Read reports and research papers • Participate in seminars and conferences • Discuss insights in meetings • Use printed reports for policy planning	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? (i.e. confusion, uncertainty, curiosity before; clarity, confidence, satisfaction after) • Before: Confusion, uncertainty, lack of insights, time pressure • After: Clarity, confidence, better understanding, satisfaction, informed decisions			

Project: Human Development Index (HDI) Predictor – A Machine Learning Approach

Empowering Decisions with Data, Predicting Development for a Better Tomorrow.